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Data Article

A dataset of whole-genome sequencing and genomic variants for the American Shad, *Alosa sapidissima*, from the Neuse River, North Carolina

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ABSTRACT

Genetic data are critical for informing future conservation decisions for the American Shad, *Alosa sapidissima*, an economically important fish in the eastern coast of the United States. To improve genomic resources for this species, we generated more than 230 Gbp of whole-genome sequencing data from an individual American Shad collected from a hatchery with parentage from the Neuse River in North Carolina, USA. These data included 94.7 Gbp from a paired-end 500-bp insert library, 68.5 Gbp from a 5-7 Kbp insert mate-pair library, and 67.5 Gbp from a 10-12 Kbp insert mate-pair library, all of which were sequenced on an Illumina NovaSeq 6000. These sequencing data were cleaned, trimmed, and aligned to an existing American Shad reference genome. The resulting variants are described, including possible effects of variants and high-impact variants by annotated gene. A total of 3.8 million variants were identified across the genome, averaging 1 variant every 237 bp. Of these variants, 64.5% were single nucleotide polymorphisms, 35.5% were insertion-deletion polymorphisms, and a total of 11.3 million variant effects were annotated. Future analyses of specific

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genes outlined by these data may be valuable in research on anadromy and life history in the American Shad and other migratory fish in addition to providing useful genomic resources for the conservation and management of this ecologically and economically important species.

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Specifications Table

Subject	Biology
Specific subject area	Genome sequencing and variation in an American Shad from Neuse River, North Carolina, USA.
Type of data	Raw, Processed. Raw genome sequencing data and processed genomic variation data with annotation.
Data collection	DNA was extracted from an American Shad using the E.Z.N.A. HP Tissue DNA Maxi Kit and whole-genome sequencing performed on a NovaSeq6000. The raw mate-pair sequences were first processed with NxTrim v0.4.3 and then all sequencing data were cleaned using fastp 0.20.0. Cleaned sequences were aligned to the American Shad reference genome using bwa v0.7.18 and processed with samtools v1.10.2. Variants were identified using bcftools 1.10.2, annotated with SnpEff v5.2e, and summarized by vcftools 0.1.17.
Data source location	Institution: University of Central Florida City/Town/Region: Orlando, Florida Country: USA Sample Origin: Edenton National Fish Hatchery, North Carolina, USA Latitude and Longitude: 36.0574 N, 76.6394 W
Data accessibility	Repository name: National Center for Biotechnology Information (NCBI) Data Identification Numbers: PRJNA487080, SRR7973879, SRR7973880, SRR7973881 Direct URLs to data: BioProject: https://www.ncbi.nlm.nih.gov/bioproject/PRJNA487080 Short insert library: https://www.ncbi.nlm.nih.gov/sra/SRR7973879 Mate pair library: https://www.ncbi.nlm.nih.gov/sra/SRR7973880 Mate pair library: https://www.ncbi.nlm.nih.gov/sra/SRR7973881 Repository name: European Variation Archive (EVA) Data Identification Number: ERZ26881722 Direct URL to data: Variant data: https://www.ebi.ac.uk/ena/browser/view/ERZ26881722
Related research article	None

1. Value of the Data

- Molecular analyses of fishes are key to understanding their life histories and population structures.
- These data provide new data on variation across the *A. sapidissima* genome.
- Generated data will contribute to studies of *A. sapidissima* genetic variation in efforts to identify appropriate broodstock and to implement effective regional conservation strategies.
- Variant data may be from an iteroparous group of American Shad, providing a template for further research on the biogeography of this species.

2. Background

American Shad (*Alosa sapidissima)* are anadromous, migratory fish endemic to the eastern United States and Canada [1]. Populations south of Cape Fear, North Carolina, USA display semel-

parity, whereas populations north of Cape Fear are iteroparous [2,3]. The disparity between these life histories and their highly heterogenous endemic range make American Shad likely to have undergone increased local adaptations [3]. Populations of American Shad have also been heavily depleted by overfishing, changing ocean conditions, and loss of freshwater spawning habitat [4]. Molecular studies have shown varying levels of population structure across their large and biogeographically diverse range [2]. The Atlantic States Marine Fisheries Commission has indicated a critical need for additional molecular genetic data for American Shad [4]. Here, we generated whole-genome sequencing data and quantified and annotated the genetic variation across the genome of American Shad from the Neuse River, North Carolina, USA. The Neuse River discharges north of Cape Fear, therefore these American Shad are representative of iteroparity as opposed to the semelparous nature of the current American Shad reference genome from St. Johns River, Florida, USA [3]. These genomic resources are valuable for understanding their life history and for the conservation of this ecologically and economically important species.

3. Data Description

One short-insert, paired-end and two long-insert, mate-pair libraries were generated from DNA extracted from an individual American Shad originating from the Neuse River, North Carolina, USA. Each library was sequenced on a NovaSeq 6000 (Illumina Inc., San Diego, CA, USA). The raw, unprocessed sequencing data are summarized in Table 1 and were deposited into the Sequence Read Archive (https://www.ncbi.nlm.nih.gov/sra/) as accessions SRR7973879–81. The entire dataset can be found in NCBI under BioProject accession PRJNA487080. More than 763 million paired-end reads representing approximately 230.6 Gbp are included in the dataset. On average, there were 254.5 (SD 51.1) million reads and 76.9 (SD 15.4) Gbp per library. In total, 93.0% and 85.5% of base pairs sequenced were of Phred-scaled quality 20 (1% error rate) or 30 (0.1% error rate) or higher and are indicative of high-quality DNA sequencing data.

Sequencing data from our North Carolina American Shad were aligned to the 903.6 Mbp reference genome (fAlsoSap.1.pri, NCBI RefSeq Assembly GCF_018492685.1) [3]. A total of 3.8 million variants were identified across the genome (averaging 1 variant per 237 bp) and were composed of 2,459,115 (64.5%) SNPs, 631,239 (16.6%) insertions, and 720,576 (18.9%) deletions. The SNPs included total of 1,943,011 transitions and 1,348,358 transversions for an overall transition/transversion ratio of 1.44. Variants were classified by SnpEff [5] into their predicted types, effects, and impacts to protein-coding genes. These predefined impact categories include “high”, “low”, “moderate”, and “modifier”. Of these, high-impact variant effects accounted for only 0.085% of all reported variants. Modifier effects accounted for more than 97% of all variant effects (Table 2). Table 3 identifies types of variant effects by count and overall percentages. The most common predicted variant effects were those found within introns (60.0%). The intronic and intergenic variants together accounted for 94.6% of variant effects. Synonymous effects (1.34%) were the most reported effects of the exonic variants, with missense (i.e., non-synonymous) effects accounting for 0.73% (Table 3). Fourteen genes were found to have at least nine high-impact variants and included *Nbeal1, Fat3a*, and *Nhsl1b* (Fig. 1).

Table 1
Summary of the raw data obtained from each DNA sequencing library. Accession numbers are for NCBI’s Sequence Read Archive (https://www.ncbi.nlm.nih.gov/sra/). Q20 and Q30 refer to the proportion of base pairs with Phred-scaled per base accuracy greater than 99% and 99.9%, respectively.

Name	Accession	Type	Insert Size	# Paired Reads	# Bases	Q20 bases	Q30 Bases
PE500	SRR7973879	151 bp; Paired-end	500 bp	313,465,270	94,666,511,540	88.2%	81.4%
MP5k	SRR7973880	151 bp; Mate-pair	5-7 Kbp	226,692,460	68,461,122,920	93.1%	85.3%
MP10k	SRR7973881	151 bp; Mate-pair	10-12 Kbp	223,358,676	67,454,320,152	92.8%	85.0%
Total	–	–	–	763,516,406	230,581,954,612	93.0%	85.5%

Table 2
The predicted impacts a variant may have on a protein-coding gene. As described by SnpEff [5] low-impact effects are assumed to have low to no impact on protein functions, moderate-impact effects may change protein efficacy, high-impact effects are predicted to result in complete or near complete loss of protein function, and modifiers include variants in mostly non-coding regions where effects are difficult to predict or assumed to be unimpactful.

Effects by Impact	Count	Percent
Low	203,004	1.805%
Moderate	92,666	0.824%
High	9,611	0.085%
Modifier	10,942,290	97.286%

Table 3
Counts and percentages of variant annotation effects categorized as by areas of effect on the genome. Original annotations provided by SnpEff [5] have been added together. A complete categorization can be found in the data section of the bioinformatics code repository (see Experimental Design, Materials, and Methods section). Note that a variant can include multiple annotations, so the sum of the annotations does not match the number of total variants identified in the genome.

Region	Type	Variant Effect	Count	Percentage
Exon	Inframe Indels	deletions	5,158	0.05%
		insertions	5,410	0.05%
	fusions	gene fusions	183	0.00%
	start	start lost	132	0.00%
		start retained	13	0.00%
		initiator	12	0.00%
	stop	stop lost	248	0.00%
		stop gained	638	0.01%
		stop retained	151	0.00%
	coding	missense	82217	0.73%
		synonymous	151071	1.34%
		frameshift	4161	0.04%
		exon loss	6	0.00%
		intragenic	17312	0.15%
	noncoding	noncoding transcript	34786	0.30%
UTR	5′	5′ variant	48,467	0.43%
		5′ truncation	2	0.00%
		5′ premature start	5,374	0.05%
	3′	3′ truncation	198,486	1.76%
		3′ variant	4	0.00%
Splice	splice	splice variant	59,224	0.52%
Intergenic	intergenic	intergenic region	1,333,729	11.80%
		upstream	1,266,334	11.20%
		downstream	1,308,237	11.57%
Intron	intronic	intron variant	6,784,697	60.01%
Total			11,300,894	100%

4. Experimental Design, Materials and Methods

The International Animal Care and Use Committee at Duke University approved all aspects of animal care, maintenance, and euthanasia (protocol #A175-15-06). Approximately 150 larval-stage *A. sapidissima* were obtained from the Edenton National Fish Hatchery (Edenton, NC, USA). The larvae were captive hatched from wild-caught adults obtained from the Neuse River in North Carolina, USA as part of a long-term restoration program for the species. The larval shad were reared to par stage (∼6 months) in a 120-cm diameter, freshwater re-circulating system at 20°C, 0 ppm salinity, and pH of 8.0. A single individual (sample ID Asap-001) was randomly collected and euthanized by immersion in MS222 followed by decapitation. DNA was extracted from 1 g of muscle tissue using the E.Z.N.A. HP Tissue DNA Maxi Kit (Omega Bio-tek, Inc., Norcross, GA, USA) following the manufacturer's recommendation. The purified DNA was submitted to the Sequencing and Genomic Technologies shared resource at the Duke Center for Genomic and Com-

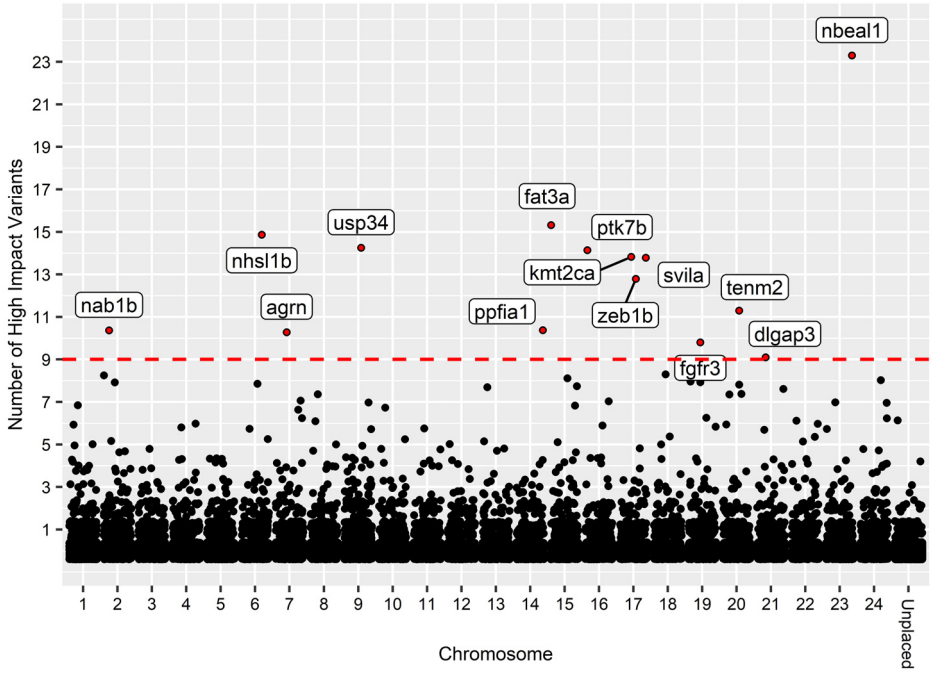

Fig. 1. Count of high-impact variants by gene. Genes with nine or more high-impact variants (denoted by the dashed red line) are labelled for visualization purposes.

putational Biology (Duke University, Durham, NC, USA) for library preparation and sequencing. One short-insert (~500 bp) library and two long-insert, mate-pair libraries (insert sizes of 5–7 Kbp and 10–12 Kbp) were generated and sequenced on an Illumina NovaSeq 6000 using 2 × 151 bp paired-end sequencing chemistry.

Reads from the two long-insert libraries were first processed with NxTrim v0.4.3 [6] to identify and split the reads at the adapter (–aggressive option), omit reads less than 50-bp in length, and convert the read pairs to the forward-reverse orientation. Next, the reads from all libraries (short-insert and long-insert) were cleaned and processed using fastp v0.20.0 [7]. The read processing included removal of sequencing adapters, bases with a quality [Q] score ≤20 from both the 5' and 3' ends, bases from the 3' end using a 4-bp sliding window and mean Q ≤20, reads with >30% uncalled (N) bases, reads with a mean Q <20, reads with a complexity score <30, and reads with a final length <50 bp. The American Shad reference genome (GCF_018492685.1) [4] was indexed with bwa v0.7.18 [8] and the cleaned sequencing reads were aligned to the reference genome using bwa mem [9]. We used samtools v1.10.2 [10] to sort the aligned reads, remove duplicates, and merge the aligned reads from each library into a single bam file. Variants were identified with bcftools v1.10.2 [10] while only retaining reads with a mapping quality ≥20 and bases with a quality ≥20. The initial set of variants were filtered to retain variants with a quality score ≥30, mean mapping quality >30, mapping quality bias Z-score ≥-9, read position bias Z-score between -5 and 5, base quality bias Z-score ≥-5, and a minimum read of 2 for the minor allele. Additionally, variants on the mitochondrial sequence (NC_014690.1) and within 10 bp of another variant were omitted.

The resulting set of filtered variants were annotated with by SnpEff v5.2e [5] and the final annotated variant file (VCF format) was uploaded to the European Variation Archive (EVA) using eva v0.4.7 [11]. Summary statistics of variants, the transition:transversion ratio, and SNP density across the genome (20 kb non-overlapping windows) were calculated using vcftools v0.1.17 [12].

Data file management and visualizations were performed in R v4.4.3 using the tidyverse package [13], ggplot2 [14], ggpp [15], ggrepel [16], grid [17], and vcfR [18]. All bioinformatics code is documented and can be accessed in GitHub (https://github.com/ea227/A.-sapidissima-Scripts).

Limitations

The authors advise that this is not a comprehensive list of all variants within the American Shad population but rather represent variation from a single diploid individual.

Ethics Statement

All procedures involving the use of animals were performed in accordance with protocol #A175-15-06 approved by the International Animal Care and Use Committee at Duke University.

Credit Author Statement

Ella S. Adams-Herrmann: Methodology, Software, Formal analysis, Data Curation, Writing - Original Draft, Visualization, Writing - Reviewing & Editing. **Lauren Currier**: Methodology, Validation, Formal analysis, Writing - Original Draft, Writing - Review & Editing, Visualization. **Tara Morgan:** Methodology, Formal analysis, Data Curation, Writing - Original Draft, Visualization. **Sönke Johnsen:** Conceptualization, Project administration, Writing - Review & Editing. **Benjamin R. Wheeler:** Conceptualization, Methodology, Investigation, Writing - Review & Editing. **Robert R. Fitak:** Conceptualization, Methodology, Investigation, Formal analysis, Data Curation, Writing - Review & Editing, Supervision, Funding acquisition.

Data Availability

ERZ26881722 (Original data) (European Variant Archive).

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Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could influence the work reported in this paper.

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